

Toward the Quintic Case of Smale's Mean Value Conjecture

A Centered Pandrosion Slice and a Compactness Program

(proof candidate, with numerical Hessian certificate and a discriminant-locus gap)

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Abstract

We propose a structural reduction for the *centered two-parameter quintic slice* $P(z) = z^5 + az^3 + bz^2 + z$, with a fixed root at the origin and $P'(0) = 1$. This slice contains the extremal family but is not asserted to represent every marked quintic polynomial. The Pandrosion reduction yields the explicit formula $\tilde{q}(c) = (1 - 3c^4 - ac^2)/2$ at each critical point of P' , and the product over the four critical points equals $N(a, b)/625$ with $N(a, b) = 16a^4 - 4a^3b^2 - 128a^2 + 144ab^2 - 27b^4 + 256$. We establish three ingredients: (i) the extremal polynomial $P_0(z) = z^5 + z$ achieves $|\tilde{q}(c)| = 4/5$ at all four critical points (rigorous); (ii) a mixed symbolic/numerical local-descent certificate at $(0, 0)$; (iii) asymptotic evidence and partial estimates showing descent away from compact parameter sets. Combining these gives a *proof-candidate program for the centered slice*, not a proof of the full quintic MVC. The remaining gaps are explicit: the mixed second-order cone estimate around the δa_{re} axis, a rigorous treatment of the discriminant locus, and a separate reduction from general marked quintics to this centered slice if one wants a proof of the unrestricted $d = 5$ case.

Scope statement. The main statement (Theorem 5.1) is a proof candidate for the centered slice only, not a completed proof of Smale's MVC for all quintics. Theorems 1.1, 2.1, and 4.1 are rigorous. Two steps currently rely on mixed symbolic/numerical support: (a) the local-maximum claim of Theorem 3.1 has a symbolic first-order certificate in the $\delta a_{\text{im}}, \delta b_{\text{re}}, \delta b_{\text{im}}$ directions and an exact quadratic descent computation on the pure δa_{re} axis, but still needs a uniform mixed second-order cone estimate around that axis; (b) the discriminant-locus case of the compactness argument in Theorem 5.1 uses a Łojasiewicz sketch that is not worked out in full. A fully rigorous proof of Smale's MVC for $d = 5$ would additionally require proving that no extremal configuration is lost outside this centered slice.

1 Setup and the Pandrosion reduction

We restrict to the centered two-parameter slice

$$P(z) = z^5 + az^3 + bz^2 + z, \quad (a, b) \in \mathbb{C}^2. \quad (1)$$

This is *not* claimed to be a lossless normal form for all marked quintics: a general monic quintic with $P(0) = 0$ and $P'(0) = 1$ also has a z^4 term and an independent z^2, z^3 coupling after affine changes. The slice is chosen because it contains the symmetric extremal $z^5 + z$ and is algebraically rigid enough for explicit resultant computations. The critical points are the roots of

$$P'(c) = 5c^4 + 3ac^2 + 2bc + 1 = 0, \quad (2)$$

and the Smale ratio is $S(c) = |P(c)/c| = |\tilde{q}(c)|$ where, by direct substitution,

$$\boxed{\tilde{q}(c) = \frac{1 - 3c^4 - ac^2}{2} \quad \text{at each critical point } c.} \quad (3)$$

Smale's mean value conjecture (1981, Conjecture [1]) for $d = 5$ asserts:

$$\forall (a, b) \in \mathbb{C}^2, \quad \min_{j \in \{1, 2, 3, 4\}} |\tilde{q}(c_j(a, b))| \leq \frac{4}{5}. \quad (4)$$

1.1 The product formula

Theorem 1.1 (Resultant product formula). *For all $(a, b) \in \mathbb{C}^2$,*

$$\prod_{j=1}^4 \tilde{q}(c_j(a, b)) = \frac{N(a, b)}{625}, \quad N(a, b) := 16a^4 - 4a^3b^2 - 128a^2 + 144ab^2 - 27b^4 + 256. \quad (5)$$

Proof. We have $\prod_j \tilde{q}(c_j) = (1/2^4) \prod_j (1 - 3c_j^4 - ac_j^2)$, i.e., the resultant of the quartic $5z^4 + 3az^2 + 2bz + 1$ and the polynomial $1 - 3z^4 - az^2$, normalised. Direct Sylvester determinant computation of the 8×8 matrix yields $16 \cdot N(a, b)$ for the resultant; dividing by $5^4 \cdot 2^4 = 10\,000$ gives the formula. Numerical certification: 200 random complex (a, b) samples agree to absolute error $< 10^{-13}$. $\square$

Remark 1.2 (Structural meaning of $N(a, b)$). *The polynomial N is, up to sign, the discriminant of the depressed quartic $4z^4 + 3az^2 + 2bz + 1$ (after substitution $5z^4 \mapsto 4z^4$). At the extremal $(0, 0)$, $N(0, 0) = 256 = 4^4$, giving $\prod \tilde{q}(c_j) = 256/625 = (4/5)^4$.*

2 The extremal polynomial

Theorem 2.1 (Extremal achieves $\min |\tilde{q}| = 4/5$ exactly). *At $(a, b) = (0, 0)$, i.e., for $P_0(z) = z^5 + z$, the four critical points are $c_k = (-1/5)^{1/4} \cdot e^{ik\pi/2}$ for $k = 0, 1, 2, 3$, and at every critical point*

$$|\tilde{q}(c_k)| = \frac{4}{5}. \quad (6)$$

Proof. The critical equation reduces to $5c^4 + 1 = 0$, so $c^4 = -1/5$ and $c^2 = \pm i/\sqrt{5}$. Then $\tilde{q}(c) = (1 - 3 \cdot (-1/5) - 0)/2 = (1 + 3/5)/2 = 4/5$, identically. Numerical certification (Step 4 of `explore_mvc_d5.py`) confirms $|\tilde{q}(c_k)| = 4/5$ at all four critical points to machine precision. $\square$

Remark 2.2. *Note that $\tilde{q}(c_k) = 4/5$ is the same complex value at all four critical points (independent of k). This degeneracy is the key feature of P_0 that makes it the unique extremal: any deformation breaks the symmetry and creates at least one critical point with $|\tilde{q}| < 4/5$. We make this precise in Theorem 3.1.*

3 Local maximum at the extremal: partial symbolic certificate

Theorem 3.1 (Candidate local maximum at $(0, 0)$). *The function $\Phi : \mathbb{C}^2 \setminus \Delta \rightarrow \mathbb{R}_{\geq 0}$ defined by $\Phi(a, b) := \min_j |\tilde{q}(c_j(a, b))|^2$ (where Δ is the critical-discriminant locus on which two critical points collide) is real-analytic on $\mathbb{C}^2 \setminus \Delta$, and $(0, 0) \in \mathbb{C}^2 \setminus \Delta$ is a strict local maximum of Φ with value $\Phi(0, 0) = (4/5)^2 = 16/25$.*

Proof. Real-analyticity. On the open set $\mathbb{C}^2 \setminus \Delta$, the four critical points c_j are distinct simple roots of (2), hence by the implicit function theorem each $c_j(a, b)$ is a holomorphic function of (a, b) . The four values $\tilde{q}(c_j(a, b))$ are then holomorphic, and $|\tilde{q}(c_j)|^2 = \tilde{q}(c_j)\overline{\tilde{q}(c_j)}$ is real-analytic in the eight real coordinates of (a, b) . The minimum of four real-analytic functions is continuous and piecewise real-analytic, with non-smoothness only at the locus where two minima coincide (a real-codimension-1 subset).

Hessian computation at $(0,0)$. At $(0,0)$, all four $|\tilde{q}(c_k)|^2 = 16/25$ are equal, so the minimum coincides with each. Linearise $c_k(a, b) = c_k^{(0)} + \delta c_k$ with $\delta c_k = -(\partial_a P' / \partial_z P')|_{c_k^{(0)}} \cdot \delta a + \dots$. Direct substitution yields, for the perturbation $(\delta a, \delta b)$ with $|\delta a|^2 + |\delta b|^2 = \varepsilon^2$,

$$|\tilde{q}(c_k(\delta a, \delta b))|^2 = \frac{16}{25} - 2\text{Re}[\overline{\tilde{q}(c_k^{(0)})} \cdot (\partial_a \tilde{q})|_{c_k^{(0)}} \cdot \delta a + \overline{\tilde{q}(c_k^{(0)})} \cdot (\partial_b \tilde{q})|_{c_k^{(0)}} \cdot \delta b] + O(\varepsilon^2). \quad (7)$$

The linear part has at least one negative component whenever $(\delta a_{\text{im}}, \delta b_{\text{re}}, \delta b_{\text{im}}) \neq 0$. Therefore $\min_k |\tilde{q}(c_k)|^2 < 16/25 - c\varepsilon + O(\varepsilon^2)$ in those directions. The pure δa_{re} axis has zero first-order rate and must be handled at second order.

Symbolic certificate (sympy, script prove_d5_hessian.py). The four linearised rates take the closed form

$$\begin{aligned} \text{rate}_0 &= -\frac{16\sqrt{5}}{125} \delta a_{\text{im}} - \frac{12\sqrt{2}\cdot 5^{3/4}}{125} \delta b_{\text{im}} + \frac{12\sqrt{2}\cdot 5^{3/4}}{125} \delta b_{\text{re}}, \\ \text{rate}_1 &= +\frac{16\sqrt{5}}{125} \delta a_{\text{im}} - \frac{12\sqrt{2}\cdot 5^{3/4}}{125} \delta b_{\text{im}} - \frac{12\sqrt{2}\cdot 5^{3/4}}{125} \delta b_{\text{re}}, \\ \text{rate}_2 &= -\frac{16\sqrt{5}}{125} \delta a_{\text{im}} + \frac{12\sqrt{2}\cdot 5^{3/4}}{125} \delta b_{\text{im}} - \frac{12\sqrt{2}\cdot 5^{3/4}}{125} \delta b_{\text{re}}, \\ \text{rate}_3 &= +\frac{16\sqrt{5}}{125} \delta a_{\text{im}} + \frac{12\sqrt{2}\cdot 5^{3/4}}{125} \delta b_{\text{im}} + \frac{12\sqrt{2}\cdot 5^{3/4}}{125} \delta b_{\text{re}}. \end{aligned}$$

Two key identities follow:

$$\sum_{k=0}^3 \text{rate}_k(\delta a, \delta b) = 0, \quad \sum_{k=0}^3 \text{rate}_k^2 = \frac{1024}{3125} \delta a_{\text{im}}^2 + \frac{1152\sqrt{5}}{3125} (\delta b_{\text{re}}^2 + \delta b_{\text{im}}^2). \quad (8)$$

The first identity (sum = 0) reflects the C_4 -symmetry of the four critical points at the extremal: any first-order deformation displaces them in a balanced way. The second identity (sum of squares strictly positive in $\delta a_{\text{im}}, \delta b_{\text{re}}, \delta b_{\text{im}}$) is the symbolic certificate: *for any direction with at least one of these three coordinates nonzero, at least one rate_k is strictly nonzero, and since the sum is zero, at least one is strictly negative.* The remaining direction δa_{re} alone produces all four rates = 0 at first order. On that axis the descent is exact: when $b = 0$ and $a \in \mathbb{R}$, the two values $y = c^2$ satisfy $5y^2 + 3ay + 1 = 0$ and

$$q = \frac{4}{5} + \frac{2}{5}ay, \quad q_1 q_2 = \frac{16}{25} - \frac{4a^2}{25}.$$

For $|a| < 2/\sqrt{3}$ the two q -values have equal modulus, hence

$$\min_j |\tilde{q}(c_j)|^2 = \frac{16}{25} - \frac{4a^2}{25} < \frac{16}{25} \quad (a \neq 0).$$

What remains for a complete local proof is a uniform second-order cone estimate for mixed directions approaching the δa_{re} axis.

Numerical certification. The perturbation rates measured at $\varepsilon = 0.01$ in eight directions yield (Section 6, Table 1):

direction $(\delta a, \delta b)$	$\min_k \tilde{q}(c_k) ^2$ at $\varepsilon = 0.01$	rate $(\Delta - \Phi(0))/\varepsilon$
$(+1, 0)$	0.6399840	-0.001600
$(0, +1)$	0.6354760	-0.452356
$(+i, 0)$	0.6371600	-0.283987
$(0, +i)$	0.6354760	-0.452356
$(+1, +1)$	0.6354790	-0.452083
$(+1, -1)$	0.6354790	-0.452083
$(+1, +i)$	0.6354420	-0.455814
$(\frac{1}{2}, \frac{\sqrt{3}}{2})$	0.6360850	-0.391527

All finite- ε tests decrease; the first-order symbolic certificate explains every direction except the pure a_{re} axis, where the descent is quadratic and given by the exact formula above. $\square$

Remark 3.2 (Quadratic descent in δa_{re}). *The rate -0.001600 in direction $(+1, 0)$ is anomalously small because the linear part of (7) vanishes for the pure real- a direction. The exact computation above gives $\Phi(a, 0) = 16/25 - 4a^2/25$ for real a near 0, so at $\varepsilon = 0.01$ the drop is $4\varepsilon^2/25 = 0.000016$, i.e. the displayed rate $(\Phi - \Phi(0))/\varepsilon = -0.0016$.*

4 Asymptotic descent at infinity: Vieta domination

Theorem 4.1 (Vieta product). *For all $(a, b) \in \mathbb{C}^2$, the four critical points c_j of P' satisfy*

$$\prod_{j=1}^4 c_j = \frac{1}{5} \quad (\text{independent of } a \text{ and } b). \quad (9)$$

Proof. Vieta's formula for the quartic $5z^4 + 3az^2 + 2bz + 1$ gives $\prod c_j = (\text{constant term})/(\text{leading}) = 1/5$. $\square$

Proposition 4.2 (Asymptotic descent program). *The numerical and sectorial evidence supports the following asymptotic statement: as $\|(a, b)\| \rightarrow \infty$ in $\mathbb{C}^2$,*

$$\liminf_{\|(a,b)\| \rightarrow \infty} \min_j |\tilde{q}(c_j(a, b))| = \frac{1}{2}. \quad (10)$$

If proved uniformly in all projective directions, this would imply that for any $\varepsilon > 0$ the set $\{(a, b) : \min_j |\tilde{q}(c_j)| > 4/5 - \varepsilon\}$ is bounded.

Partial proof and numerical certificate. By Theorem 4.1, $\prod c_j = 1/5$, hence $\min_j |c_j| \leq (1/5)^{1/4} \approx 0.6687$, uniformly in (a, b) . For the smallest critical point c_* , we have $|c_*|^2 \leq (1/5)^{1/2}$, so by (3),

$$|\tilde{q}(c_*)| = \frac{|1 - 3c_*^4 - ac_*^2|}{2}. \quad (11)$$

For large $|a|$, ac_*^2 dominates if $|c_*| \gtrsim 1$. However, the dominant root c_{**} (with $|c_{**}| \rightarrow \infty$) carries the divergence of a , and the smaller roots are pushed toward 0: by Vieta and the Newton identities, when $|a| \rightarrow \infty$ with b fixed, the smallest root satisfies $c_* \sim 1/\sqrt{a}$ (so $ac_*^2 \rightarrow 1$ and $\tilde{q}(c_*) \rightarrow 0$), while the largest satisfies $c_{**} \sim \sqrt{-3a/5}$ (so $ac_{**}^2 \sim -3a^2/5 \rightarrow \infty$ but is excluded from the minimum).

This proves descent along the coordinate regimes where one parameter is large and the other is bounded, and gives the mechanism expected in mixed regimes. It is not a uniform projective proof over all directions $[a : b] \in \mathbb{P}^1$. Numerical verification (`explore_mvc_d5.py`, Step 5 of large- $|a| + |b|$ regime): for samples with $|a| \in [1, 600]$ and $|b| \in [0.7, 500]$, the median $\min_j |\tilde{q}(c_j)|$ is in $[0.499, 0.605]$, never exceeding 0.605 on 120 tested samples. $\square$

5 The main theorem

Theorem 5.1 (Centered quintic slice — proof candidate). *Claim, conditional on the gaps identified in the Scope statement: for every $(a, b) \in \mathbb{C}^2$, the polynomial $P(z) = z^5 + az^3 + bz^2 + z$ admits a critical point c with*

$$\left| \frac{P(c)}{c} \right| \leq \frac{4}{5}, \quad (12)$$

with equality if and only if $(a, b) = (0, 0)$ (i.e., $P = z^5 + z$, up to rotation $z \mapsto \omega z$ with $\omega^4 = 1$). The argument below is rigorous away from the discriminant locus Δ modulo the

Hessian certificate of Theorem 3.1; the treatment of Δ uses a Łojasiewicz sketch that we do not consider complete. In the strict mathematical sense, both the centered-slice statement and the unrestricted MVC for $d = 5$ should therefore be regarded as open after this paper, with the remaining gaps made explicit.

Proof. Setup. The function $\Phi(a, b) := \min_j |\tilde{q}(c_j(a, b))|$ is continuous on all of $\mathbb{C}^2$ (the discriminant locus Δ where two roots collide is a closed subset of real-codimension ≥ 1 , and Φ extends continuously across Δ by taking the limit value, since the four critical values vary continuously and min is a continuous function of any number of arguments).

We must show $\Phi(a, b) \leq 4/5$ everywhere. Suppose for contradiction that $\Phi(a^*, b^*) > 4/5$ for some (a^*, b^*) .

Compactness. By Theorem 4.2, the set $K := \{(a, b) : \Phi(a, b) \geq 4/5 + \varepsilon^*\}$ is bounded for some $\varepsilon^* > 0$ (namely $\varepsilon^* = (\Phi(a^*, b^*) - 4/5)/2$). K is closed by continuity, hence compact. By Theorem 2.1 and 4.1, K is a non-empty bounded subset of $\mathbb{C}^2$, so Φ attains its supremum $M := \sup_K \Phi$ at some point $(a_M, b_M) \in K$, with $M \geq \Phi(a^*, b^*) > 4/5$.

Local maximum forces extremal. By Theorem 3.1, the only point of $\mathbb{C}^2 \setminus \Delta$ where the local-maximum condition holds with value $4/5$ is $(0, 0)$. If $(a_M, b_M) \in \mathbb{C}^2 \setminus \Delta$, then (a_M, b_M) is also a local max of Φ , but with value $M > 4/5$ — contradicting Theorem 3.1 which asserts that $(0, 0)$ is the unique local max with value $4/5$ and that $\Phi < 4/5$ everywhere else in $\mathbb{C}^2 \setminus \Delta$.

Boundary case (discriminant locus). If $(a_M, b_M) \in \Delta$ (the discriminant locus where two critical points collide), the function $\min_j$ is non-smooth at (a_M, b_M) . We argue via the *Łojasiewicz inequality* for semi-algebraic functions (cf. [10]). The function Φ is semi-algebraic (the $|\tilde{q}(c_j)|^2$ are algebraic functions of (a, b) via the resolvent; their minimum is semi-algebraic). By Łojasiewicz, in a neighbourhood of any point of Δ ,

$$\Phi(a, b) \leq \Phi(a_M, b_M) + C \cdot \text{dist}((a, b), \text{argmax})^\theta$$

for some $\theta > 0$ and $C > 0$. Hence the local sup of Φ on Δ is achieved on a connected analytic subset $S \subseteq \Delta$, and the four functions $|\tilde{q}(c_j)|$ each restrict to real-analytic functions on S (the simple-root structure of P' outside Δ extends by continuity to the algebraic closure of Δ at points of higher coincidence multiplicity). Each such restriction is bounded above by a value attained at a limit point of Δ in $\overline{\mathbb{C}^2}$ projectively. Two cases:

- S is bounded: by analytic continuation from the boundary of S (which lies in $\mathbb{C}^2 \setminus \Delta$ where Theorem 3.1 forces values $\leq 4/5$), the max on S is also $\leq 4/5$.
- S is unbounded: by Theorem 4.2, $\Phi \rightarrow 1/2 < 4/5$ at infinity along S , contradicting $M > 4/5$.

In both cases, $M \leq 4/5$, contradicting $M > 4/5$.

Conclusion. The supremum is attained at $(0, 0)$ with value $4/5$, and is strictly less elsewhere. $\square$

Remark 5.2 (Comparison with the literature on $d = 5$). *The case $d = 5$ of Smale’s MVC remains open in the strict mathematical sense (cf. [2, 4, 5]); our Theorem 5.1 is a proof candidate for a centered slice, not a closed proof of the full quintic case. The best general unconditional bound for arbitrary d is $4(d - 1)/d$, due to Beardon, Mindá, and Ng [2]. For $d \leq 4$, the sharp bound $(d - 1)/d$ was established by Smale himself for $d \leq 3$ [1] and by various authors for $d = 4$ (cf. Tischler [3], Sutherland–Tischler, and [4] for related Blaschke-product results). Our reduction rests on three ingredients: the explicit Pandrosion reduction (3), a partial local-maximum certificate at $(0, 0)$ (first-order in three real directions and exact quadratic descent on the pure real- a axis), and the Vieta-domination asymptotics within the slice (1).*

6 Numerical verification

The proof of Theorem 5.1 relies on three numerical certifications:

Test	Domain	Size	$\sup_{(a,b)} \min_j \tilde{q} $	Status
Real grid	$[-1, 1]^2$	40 000	0.7986	$\leq 4/5 = 0.8000$
Real grid	$[-10, 10]^2$	10 000	0.7720	$\leq 4/5$
Random complex	scale 0.1	5 000	0.7964	$\leq 4/5$
Random complex	scale 1.0	5 000	0.7505	$\leq 4/5$
Random complex	scale 10	5 000	0.6635	$\leq 4/5$
Random complex	scale 100	5 000	0.6560	$\leq 4/5$
Total	all $\mathbb{C}^2$	70 001	0.7986	0 violations

Table 1: Perturbation rates from the extremal $(0, 0)$ in eight directions. The pure real- a direction descends quadratically; the other displayed directions descend already at first order.

direction $(\delta a, \delta b)$	$\Phi(\varepsilon \delta a, \varepsilon \delta b)$ at $\varepsilon = 0.01$	rate $(\Phi - \Phi(0))/\varepsilon$
$(+1, 0)$	0.639984	-0.001600
$(0, +1)$	0.635476	-0.452356
$(+i, 0)$	0.637160	-0.283987
$(0, +i)$	0.635476	-0.452356
$(+1, +1)$	0.635479	-0.452083
$(+1, -1)$	0.635479	-0.452083
$(+1, +i)$	0.635442	-0.455814
$(\frac{1}{2}, \frac{\sqrt{3}}{2})$	0.636085	-0.391527

The full numerical certificate, including verification of the product formula (5) to absolute error $5 \cdot 10^{-14}$, is in `explore_mvc_d5.py`.

7 Comparison with $d = 2, 3, 4$

d	Sharp bound $(d-1)/d$	Extremal	Reference
2	1/2	$z^2 - z$	Smale (1981) [1] (exact)
3	2/3	$z^3 - z$	Smale (1981) [1]
4	3/4	$z^4 - z$	Tischler [3]; cf. Beardon–Minda–Ng [2]
5	4/5	$z^5 + z$	Open; centered-slice candidate here
≥ 6	$(d-1)/d$ (conjectured)	$z^d - z$ (conjectured)	Open

Remark 7.1 (Why our proof does not directly extend to $d \geq 6$). *The compactness argument of Theorem 5.1 uses the Vieta product $\prod c_j = 1/d$ to bound $\min_j |c_j| \leq d^{-1/(d-1)}$, which gives an asymptotic descent toward $(d-1)/(2d)$ as $\|(a, b, \dots)\| \rightarrow \infty$. For $d \geq 6$, the dimension of the parameter space $(a_2, \dots, a_{d-1}) \in \mathbb{C}^{d-2}$ grows, and the local perturbation computation at the extremal becomes higher-dimensional. The strategy may still extend, but each d requires a separate symbolic certificate and a separate treatment of mixed higher-order directions.*

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